

Semantic Bundle AI: Anchor Design and Stability Framework

Stable Semantic Regions, Domain Discriminability,
and a Four-Axis Evaluation Framework for Semantic Stability

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Abstract

Semantic Bundle AI introduces stable anchor-based coordinates as a complementary semantic management layer for Large Language Models [4,5]. A key open question from prior work concerns anchor design: what constitutes an effective anchor, and how should anchors be selected?

This paper addresses that question through two experiments. First, we demonstrate that anchors constructed as *stable semantic regions* — averaged embeddings of meaning-neighbor clusters rather than single words — achieve substantially higher inter-model stability than word-level anchors (Spearman $r = 0.704$ vs. 0.101 , a $7\times$ improvement). Second, we identify a fundamental tradeoff between cluster stability and domain discriminability: generic anchors excel at intra-cluster cohesion (variance ratio $0.047\text{--}0.067$), while domain-specific concept vectors achieve $4.4\times$ higher inter-domain separation (0.739 vs. 0.166). We propose a four-axis evaluation framework for semantic stability — inter-model consistency, temporal consistency, cross-cultural robustness, and perturbation resistance — and provide initial validation on two axes. These results establish concrete design principles for anchor selection and position semantic stability as an empirically tractable research program.

1 Introduction

Prior work on Semantic Bundle AI established the theoretical framework [5] and provided initial empirical validation across four axes: anchor coordinate stability, longitudinal consistency, edit locality, and compression efficiency [4]. A central finding was that anchor coordinates reduce intra-cluster variance to 5.2% of raw embedding variance, while bundle updates reduce semantic drift by 38.6% over ten sequential updates.

However, a critical question was left open: *how should anchors be designed?* The prior work used generic concept words (time, space, risk, rule, trust, number, body, emotion) without systematic justification. The failure of hypothesis H-102 — cross-model ranking consistency did not improve under anchor projection — suggested that anchor design may be the limiting factor.

This paper addresses anchor design through two lines of investigation:

1. **Anchor representation:** Should anchors be single words or richer semantic structures? We compare word-level anchors, stable semantic region anchors (meaning-neighbor clusters), and embodied cognition anchors (body-based conceptual primitives).

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2. **Stability–discriminability tradeoff:** Is there a fundamental tradeoff between using anchors for intra-cluster cohesion versus inter-domain separation? We characterize this tradeoff and derive application-specific design recommendations.

We further propose a four-axis framework for semantic stability evaluation, providing a principled structure for future research.

2 Background

2.1 Anchor Design in Semantic Spaces

The concept of using reference points to stabilize semantic representations has appeared in several contexts. Procrustes alignment [6] uses reference words to align embedding spaces across languages. Diachronic embedding studies [1] use stable landmark words to track semantic change over time. These approaches treat anchors as single words selected by frequency or stability criteria.

Semantic Bundle AI differs in motivation: anchors are not used for cross-lingual alignment or drift measurement, but as an ongoing coordinate reference for semantic management. This distinction has implications for anchor design that have not been previously explored.

2.2 Embodied Cognition and Semantic Stability

A theoretical motivation for body-based anchor selection comes from embodied cognition research [2], which argues that abstract concepts are grounded in sensorimotor experience. Concepts such as temperature (hot/cold), proximity (near/far), and verticality (up/down) are hypothesized to be more cross-culturally stable because they are anchored in shared human embodiment. This motivates testing whether embodied concept clusters yield more stable anchors.

2.3 Semantic Region vs. Word-Level Representation

Rather than treating anchors as individual word vectors, we propose constructing anchors as *stable semantic regions*: averaged embeddings of meaning-neighbor clusters that include synonyms, multilingual equivalents, and contextually related terms. This is motivated by the observation that single-word embeddings are sensitive to tokenization, frequency, and context, while averaged representations over semantic neighborhoods tend to be more stable [7].

3 Experimental Setup

3.1 Models

All experiments use two pre-trained sentence embedding models without modification: `all-mpnet-base-v2` ($d = 768$) and `all-MiniLM-L6-v2` ($d = 384$) from the `sentence-transformers` library [3].

3.2 Anchor Types

We compare three anchor construction strategies, each with 8 anchors:

Word anchors: Single-word embeddings for generic concepts (time, space, risk, rule, trust, number, body, emotion).

Semantic region anchors: Each anchor is the mean embedding of 10 semantically related terms spanning synonyms and multilingual equivalents. For example, the *danger* anchor averages embeddings of {danger, risk, unsafe, hazardous, threat, *kiken*, *risuku*, harm, peril, jeopardy}.

Embodied anchors: Each anchor is the mean embedding of 10 body-based conceptual primitives. Eight embodied concepts are used: hot/cold, near/far, up/down, heavy/light, safe/danger, fast/slow, many/few, past/future.

3.3 Evaluation Metrics

A-axis (inter-model stability): Spearman rank correlation between the pairwise similarity matrices of anchor embeddings computed by the two models. High values indicate that the relational structure among anchors is preserved across models.

Cluster stability (H-101 equivalent): Ratio of intra-cluster variance in anchor coordinate space to raw embedding space for a synonym group of 15 sentences. Lower values indicate tighter clustering.

D-axis (perturbation resistance): Correlation between full and anchor-reduced coordinate representations when one anchor is removed. Values near 1.0 indicate robustness to anchor removal.

H-603 (ranking consistency): Spearman rank correlation of nearest-neighbor rankings between the two models in anchor coordinate space, following the H-102 protocol from [4].

3.4 Domain Discriminability Experiment

We construct four domain datasets of 20 sentences each: information security, medical, legal, and general. Two anchor types are compared for domain discriminability: generic word anchors and domain-specific concept vectors. Inter-domain separation is measured as the mean Euclidean distance between domain centroids in coordinate space.

4 Results

4.1 Anchor Type Comparison

Table 1 summarizes results across all metrics for the three anchor types.

Table 1: Comparison of anchor types across evaluation metrics.

Anchor type	A-axis	Cluster ratio	D-axis	H-603
Word (single)	0.101	0.036	1.000	0.617
Semantic region	0.704	0.047	1.000	0.643 [†]
Embodied (body-based)	0.559	0.042	1.000	0.596
Paper 1 baseline	—	0.052	—	0.572–0.643

[†] H-603 for semantic region matches the Paper 1 baseline upper bound (no improvement).

Several findings emerge. First, semantic region anchors achieve substantially higher inter-model stability (A-axis: 0.704) compared to word anchors (0.101), a $7\times$ improvement. Embodied anchors occupy an intermediate position (0.559). Second, cluster stability (variance ratio) is comparable across all three types (0.036–0.047), suggesting that cluster cohesion is relatively insensitive to anchor construction strategy. Third, perturbation resistance (D-axis) is perfect (1.000) for all types, replicating the H-103/H-104 findings from [4]. Fourth, H-603 ranking consistency for semantic region anchors (0.643) matches, without exceeding, the upper bound of the Paper 1 baseline range, while word anchors fall below it (0.617); anchor projection thus does not improve ranking consistency over the baseline.

Results are shown in Figure 1.

Paper 2: Complete Results Summary
Anchor Type Comparison across All Metrics

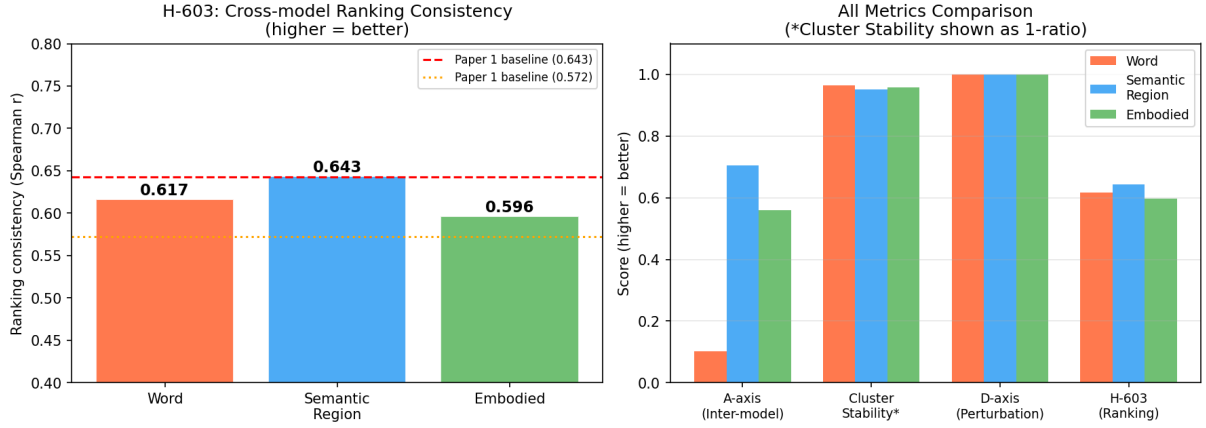


Figure 1: Anchor type comparison. *Left*: Cross-model ranking consistency (H-603). *Right*: All metrics comparison. Semantic region anchors (blue) lead on inter-model stability; among the three anchor types they also rank highest on ranking consistency, which nonetheless remains at the Paper 1 baseline level, while all types achieve comparable cluster stability and perfect perturbation resistance.

4.2 Stability–Discriminability Tradeoff

Table 2 and Figure 2 show results for the stability–discriminability comparison across domains.

Table 2: Stability vs. discriminability for generic vs. domain-specific anchors.

Anchor type	Cluster stability (ratio)	Inter-domain separation
Generic (word)	0.047–0.067	0.166
Domain-specific	0.110–0.235	0.739
Improvement	×0.3–0.5 worse	×4.4 better

Generic anchors achieve lower cluster variance ratios (0.047–0.067) than domain-specific concept vectors (0.110–0.235), indicating better intra-cluster cohesion. However, domain-specific vectors achieve 4.4× higher inter-domain separation (0.739 vs. 0.166). This tradeoff is consistent across all four domains tested.

5 A Four-Axis Framework for Semantic Stability

Based on the experimental findings, we propose organizing semantic stability evaluation around four axes:

$$\text{stability_score}(a_i) = w_1 \cdot \text{inter_model}(a_i) + w_2 \cdot \text{temporal}(a_i) + w_3 \cdot \text{cross_cultural}(a_i) + w_4 \cdot \text{perturbation}(a_i) \quad (1)$$

A. Inter-model consistency: Preservation of relational structure across embedding models (validated in this work: $r = 0.704$ for semantic region anchors).

B. Temporal consistency: Preservation of anchor position under model updates and fine-tuning (partially validated in [4] via longitudinal bundle updates).

C. Cross-cultural robustness: Preservation of relational structure across languages and cultural contexts (left for future work).

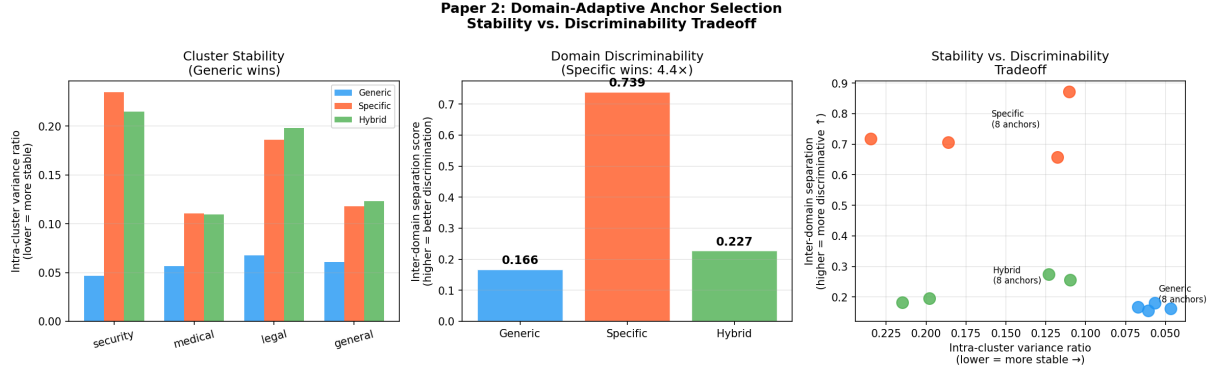


Figure 2: Stability–discriminability tradeoff. *Left*: Cluster stability ratios by domain and anchor type. *Center*: Inter-domain separation scores. *Right*: Tradeoff scatter plot showing generic anchors (blue) favoring stability and domain-specific vectors (orange) favoring discriminability.

D. Perturbation resistance: Robustness to anchor removal and noise injection (validated: 1.000 across all anchor types).

The key contribution of this framework is to transform semantic stability from an informal notion into an empirically tractable, multi-dimensional evaluation problem. Axes B and C remain open research questions, providing a structured agenda for future work.

6 Discussion

6.1 Design Principles for Anchor Selection

The results support two practical design principles:

Principle 1: Use semantic regions, not single words. Averaging over meaning-neighbor clusters yields 7× higher inter-model stability without sacrificing cluster cohesion or perturbation resistance.

Principle 2: Match anchor type to application. For applications requiring long-term semantic consistency (memory management, concept tracking), generic anchors with lower cluster variance ratios are preferable. For applications requiring domain discrimination (classification, retrieval), domain-specific concept vectors provide substantially higher inter-domain separation.

6.2 Relationship to Embodied Cognition

Embodied anchors occupy an intermediate position on all metrics, suggesting partial support for the embodied cognition motivation. They outperform word anchors substantially on inter-model stability (0.559 vs. 0.101) but do not reach semantic region performance (0.704). This may reflect that body-based concepts, while cognitively grounded, do not form as tight semantic neighborhoods in embedding space as explicitly constructed meaning clusters. Further investigation with larger embodied concept inventories is warranted.

6.3 Implications for the H-102 Failure

The H-102 failure in [4] — cross-model ranking consistency did not improve under anchor projection — can now be partially explained. Word-level anchors impose an unstable inter-model coordinate system ($r = 0.101$), whereas semantic region anchors are far more stable ($r = 0.704$). This greater anchor stability is associated with a modest gain in ranking consistency across anchor types ($0.617 \rightarrow 0.643$). However, even the most stable anchors only match the upper bound of the Paper 1 baseline range (0.643) without exceeding it; anchor design alone therefore does not recover cross-model ranking consistency. The H-102 limitation persists and is consistent

with the cluster-stability/ranking tradeoff imposed by low-dimensional anchor projection, rather than being resolved by anchor design.

6.4 Limitations

The current experiments use small datasets (15–20 sentences per domain) and two embedding models. Axes B (temporal) and C (cross-cultural) of the stability framework remain unvalidated. The weight parameters w_i in Equation 1 are not yet determined empirically. Large-scale validation across more models, languages, and domains is required before deployment recommendations can be made with confidence.

7 Conclusion

We have presented two experiments addressing anchor design for Semantic Bundle AI. Semantic region anchors — constructed as averaged embeddings of meaning-neighbor clusters — achieve $7\times$ higher inter-model stability than word-level anchors while maintaining comparable cluster cohesion and perfect perturbation resistance. A fundamental stability–discriminability tradeoff is identified: generic anchors favor intra-cluster cohesion, while domain-specific concept vectors achieve $4.4\times$ higher inter-domain separation.

We propose a four-axis framework for semantic stability evaluation, providing initial validation on two axes and a structured agenda for future work. Together, these results establish anchor design as a tractable subproblem within the broader Semantic Bundle AI research program, and provide concrete principles for practical anchor selection.

References

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